

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- ✓ LATEST NTA PATTERN
- ✓ CONCEPT VIDEOS
- ✓ LIVE DOUBT SESSIONS
- ✓ DPP, REVISION SHEETS
- ✓ 100% MONEYBACK TO MERITORIOUS STUDENTS

JOIN NOW

YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir



JEE ADV. CRASH COURSE

TARGETING JEE ADVANCED 2020

- ✓ CONCEPT BUILDING VIDEOS
- ✓ ADVANCED LEVEL VIDEOS
- ✓ 2 LEVELS OF SHEETS,
1 WITH ALL TYPE OF QUES.
- ✓ LIVE DISCUSSION OF
PREVIOUS YEAR QUESTIONS

JOIN NOW

YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir



JEE Mains 2020 Jan Chapter wise Question Bank

Probability

Q1

An unbiased coin is thrown 5 times. Let X be a random variable and k be the value assigned to X for $k = 3, 4, 5$ times Head occurs consecutively and otherwise the value of X is assigned -1 . What is value of expectation.

एक निष्पक्षपाती सिक्के को 5 बार उछाला जाता है माना X यादृच्छिक चर है तथा X से अर्जित मान k है जहां $k = 3, 4, 5$ बार लगातार शीर्ष आता है अन्यथा X को -1 मान दिया जाता है तब इसकी आकांक्षा है—

(1) $\frac{1}{8}$

(2) $-\frac{1}{8}$

(3) $\frac{3}{8}$

(4) $-\frac{3}{8}$

7th Jan Morning

Sol

(1)

k	0	1	2	3	4	5
P(k)	$\frac{1}{32}$	$\frac{12}{32}$	$\frac{11}{32}$	$\frac{5}{32}$	$\frac{2}{32}$	$\frac{1}{32}$

k = no. of times head occur consecutively

k = शीर्ष के क्रमागत आने की संख्या

Now expectation अब आकांक्षा

$$= \sum xP(k) = (-1) \times \frac{1}{32} + (-1) \times \frac{12}{32} + (-1) \times \frac{11}{32} + 3 \times \frac{5}{32} + 4 \times \frac{2}{32} + 5 \times \frac{1}{32} = \frac{1}{8}$$

02

There are 5 machines. Probability of a machine being faulted is $\frac{1}{4}$. Probability of atmost two machines

is faulted, is $\left(\frac{3}{4}\right)^3$ k then value of k is

5 मशीन इस प्रकार है कि किसी एक मशीन के खराब होने की प्रायिकता $\frac{1}{4}$ है, यदि अधिकतम दो मशीनों के खराब होने

की प्रायिकता $\left(\frac{3}{4}\right)^3$ k है, तो k का मान होगा।

(1) $\frac{17}{2}$

(2) 4

(3) $\frac{17}{8}$

(4) $\frac{17}{4}$

Sol

(3)

Required probability = when no. machine has fault + when only one machine has fault + when only two machines have fault.

अभीष्ट प्रायिकता = जब कोई भी मशीन खराब नहीं है + जब केवल एक मशीन खराब है + जब केवल दो मशीन खराब है

$$\begin{aligned}
 &= {}^5C_0 \left(\frac{3}{4}\right)^5 + {}^5C_1 \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)^4 + {}^5C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^3 \\
 &= \frac{243}{1024} + \frac{405}{1024} + \frac{270}{1024} = \frac{918}{1024} = \frac{459}{512} = \frac{27 \times 17}{64 \times 8} \\
 &= \left(\frac{3}{4}\right)^3 \times k = \left(\frac{3}{4}\right)^3 \times \frac{17}{8} \\
 \therefore k &= \frac{17}{8}
 \end{aligned}$$

03

Let $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{6}$ where A and B are independent events then

माना $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{6}$ जहाँ A तथा B स्वतंत्र घटनाएँ हैं, तब

$$(1) P\left(\frac{A}{B}\right) = \frac{1}{6} \quad (2) P\left(\frac{A}{B'}\right) = \frac{1}{3} \quad (3) P\left(\frac{A}{B'}\right) = \frac{2}{3} \quad (4) P\left(\frac{A}{B}\right) = \frac{5}{6}$$

Sol

(2)

A & B are independent events so

A तथा B स्वतंत्र घटनाएँ हैं, इसलिए

$$P\left(\frac{A}{B'}\right) = \frac{1}{3}$$

04

Let A and B are two events such that $P(\text{exactly one}) = \frac{2}{5}$, $P(A \cup B) = \frac{1}{2}$ then $P(A \cap B) =$

माना A और B दो घटनाएँ इस प्रकार हैं कि $P(\text{ठीक एक}) = \frac{2}{5}$, $P(A \cup B) = \frac{1}{2}$ तब $P(A \cap B) =$

$$(1) \frac{1}{10} \quad (2) \frac{2}{9} \quad (3) \frac{1}{8} \quad (4) \frac{1}{12}$$

Sol

(1)

$$P(\text{exactly one ठीक एक}) = \frac{2}{5}$$

$$\Rightarrow P(A) + P(B) - 2P(A \cap B) = \frac{2}{5}$$

$$P(A \cup B) = \frac{1}{2}$$

$$\Rightarrow P(A) + P(B) - P(A \cap B) = \frac{1}{2}$$

$$\therefore P(A \cap B) = \frac{1}{2} - \frac{2}{5} = \frac{5-4}{10} = \frac{1}{10}$$

05

In a bag there are 20 cards 10 names A and another 10 names B. Cards are drawn randomly one by one with replacement then find probability that second A comes before third B.

एक थैले में 20 पत्ते हैं जिनमें से 10 पर A तथा 10 पर B अंकित हैं यदि पत्तों को एक-एक कर पुनर्वापति के साथ निकाला जाता है तो तीसरे B से पहले दूसरा A प्राप्त होने की प्रायिकता है—

(1) $\frac{13}{16}$

(2) $\frac{11}{16}$

(3) $\frac{7}{16}$

(4) $\frac{9}{16}$

Sol

(2)

$$AA + ABA + BAA + ABBA + BBAA + BABA$$

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{11}{16}$$

06

Let probability distribution is

माना प्रायिकता वितरण इस प्रकार है

$x_i :$	1	2	3	4	5
$P_i :$	k^2	$2k$	k	$2k$	$5k^2$

then value of $p(x > 2)$ is

तब $p(x > 2)$ का मान है

(1) $\frac{7}{12}$

(2) $\frac{1}{36}$

(3) $\frac{1}{6}$

(4) $\frac{23}{36}$

Sol

(4)

$$\sum p_i = 1 \Rightarrow 6k^2 + 5k = 1$$

$$6k^2 + 5k - 1 = 0$$

$$6k^2 + 6k - k - 1 = 0$$

$$(6k - 1)(k + 1) = 0 \Rightarrow k = -1 \text{ (rejected असमान्य) ; } k = \frac{1}{6}$$

$$P(x > 2) = k + 2k + 5k^2$$

$$= \frac{1}{6} + \frac{2}{6} + \frac{5}{36} = \frac{6+12+5}{36} = \frac{23}{36}$$



mathongo

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- ✓ LATEST NTA PATTERN
- ✓ CONCEPT VIDEOS
- ✓ LIVE DOUBT SESSIONS
- ✓ DPP, REVISION SHEETS
- ✓ 100% MONEYBACK TO MERITORIOUS STUDENTS

JOIN NOW

YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir



JEE ADV. CRASH COURSE

TARGETING JEE ADVANCED 2020

- ✓ CONCEPT BUILDING VIDEOS
- ✓ ADVANCED LEVEL VIDEOS
- ✓ 2 LEVELS OF SHEETS,
1 WITH ALL TYPE OF QUES.
- ✓ LIVE DISCUSSION OF
PREVIOUS YEAR QUESTIONS

JOIN NOW

YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir

